

SchooliEdu

Backend API Guide

Industry-style reference for API testing, payloads, permissions, and database impact.

Version 0.2 | Generated 2026-06-06 10:29

Document Scope

- Documents all APIs currently implemented in the backend.
- Covers endpoint purpose, permission, request payload or query parameters, expected response, and affected tables.
- Separates implemented APIs from schema-only roadmap areas.
- Designed to be used alongside Swagger UI while manually testing the backend.

Quick Start

- Run the backend, then open <http://localhost:5000/api/docs>.
- Call POST `/api/auth/login` using admin credentials.
- Copy `data.accessToken` into Swagger Authorize as a bearer token.
- Create teachers/students, assign them, add teacher availability, then schedule classes.

Usage Conventions

Base URL: `http://localhost:5000`

Authentication: protected APIs require Authorization: Bearer <accessToken>.

Response envelope: successful JSON responses use success, message, data, and optional pagination. Errors use success=false with error.code and optional error.details.

Dates: use ISO 8601 date-time strings for timestamps. The project default timezone for scheduling is Asia/Kolkata.

Permission Matrix

Role	Operational scope
Admin	Full seeded permission set; can create users, schedule/cancel/reschedule classes, override conflicts, mark/override attendance, manage settings.
Teacher	View assigned students/classes, join class, mark attendance, create/review homework later, view reports.
Student	View own classes, join class, view own attendance, submit homework later, create tickets, view certificates/reports.
Support	Operational view of users/classes/attendance/credits/tickets/reports; can resolve tickets and assist rescheduling.

Database Impact Map

Area	Tables
Identity and RBAC	users, roles, permissions, user_roles, role_permissions, refresh_tokens, password_reset_tokens, otp_tokens
People and scheduling prerequisites	teacher_student_assignments, teacher_availability, teacher_unavailable_dates
Class execution	classes, class_participants, class_attendance
Zoom evidence	zoom_meetings, zoom_attendance_events
Cancellation workflow	class_cancellation_requests
Notification configuration	email_templates, notification_rules, notification_delivery_logs
Planned API schemas	homework, homework_resources, homework_submissions, homework_submission_files, credits_ledger, tickets, ticket_messages, ticket_attachments, ticket_status_history, certificates, reports, system_settings, audit_logs

Recommended Manual Testing Sequence

- 1. Login and authorize Swagger.
- 2. Create or verify a teacher user and a student user.
- 3. Create an active teacher-student assignment.
- 4. Add teacher weekly availability.
- 5. Run class conflict check.
- 6. Schedule class and confirm Zoom metadata.
- 7. Use join endpoint to fetch student/teacher meeting links.
- 8. Create and review a cancellation request.
- 9. Mark attendance and verify participant attendance status sync.
- 10. Create email templates and notification rules.

Platform Entry Points

Use these endpoints to confirm the service is available and to access the interactive API explorer.

Endpoint Catalogue

Method	Path	Permission
GET	/api/health	Public
GET	/api/docs	Public
GET	/api/docs.json	Public

Detailed Endpoint Specs

GET

/api/health

Purpose

Backend liveness check.

Auth

Public

Response

Returns service name, environment, and timestamp.

Tables

No table impact.

Request

No payload.

GET

/api/docs

Purpose

Swagger UI for manual testing.

Auth

Public

Response

HTML Swagger UI.

Tables

No table impact.

Notes

Use Authorize with a bearer token after login.

Request

No payload.

GET

/api/docs.json

Purpose

Raw OpenAPI specification.

Auth

Public

Response

OpenAPI JSON.

Tables

No table impact.

Request

No payload.

Authentication

Authenticate users and manage access/refresh token lifecycle.

Endpoint Catalogue

Method	Path	Permission
POST	/api/auth/login	Public
POST	/api/auth/refresh	Public
POST	/api/auth/logout	Public
GET	/api/auth/me	Bearer token

Detailed Endpoint Specs

POST

/api/auth/login

Purpose

Login with username/email and password.

Auth

Public

Response

Returns accessToken, refreshToken, and user profile.

Tables

Reads users, roles, permissions. Inserts refresh_tokens.

Notes

Use identifier as either username or email.

Request

```
{ "identifier": "admin", "password": "Schooli@2025" }
```

POST

/api/auth/refresh

Purpose

Issue a fresh access token from a refresh token.

Auth

Public

Response

Returns a new access token payload.

Tables

Reads refresh_tokens and users.

Request

```
{ "refreshToken": "<refresh token>" }
```

POST

/api/auth/logout

Purpose

Revoke a refresh token.

Auth

Public

Response

Returns logout confirmation.

Tables

Updates refresh_tokens.

Request

```
{ "refreshToken": "<refresh token>" }
```

GET**/api/auth/me****Purpose** Fetch current authenticated user context.**Auth** Bearer token**Response** Returns profile, roles, and permissions.**Tables** Reads users, roles, permissions.**Request**

No payload.

RBAC

Inspect roles and permissions used by route guards.

Endpoint Catalogue

Method	Path	Permission
GET	/api/roles	role.view
GET	/api/permissions	permission.view

Detailed Endpoint Specs

GET /api/roles	
Purpose	List all platform roles.
Auth	role.view
Response	Returns roles.
Tables	Reads roles.
Request	
No payload.	

GET /api/permissions	
Purpose	List all permission keys.
Auth	permission.view
Response	Returns permissions.
Tables	Reads permissions.
Request	
No payload.	

User Administration

Create and manage base accounts for admins, teachers, students, and support users.

Endpoint Catalogue

Method	Path	Permission
GET	/api/users	user.view
POST	/api/users	user.create
GET	/api/users/{id}	user.view
PATCH	/api/users/{id}	user.update
PATCH	/api/users/{id}/status	user.deactivate
POST	/api/users/{id}/roles	user.update

Detailed Endpoint Specs

GET

/api/users

- Purpose

List users with filters.
- Auth

user.view
- Response

Paginated user list.
- Tables

Reads users, user_roles, roles.
- Request

Query: page, limit, search, role, status.

POST

/api/users

- Purpose

Create user and optional roles.
- Auth

user.create
- Response

Created user profile.
- Tables

Inserts users and user_roles.
- Notes

Passwords are hashed before storage.
- Request

{ "firstName": "Demo", "lastName": "Teacher", "username": "demo_teacher", "email": "demo.teacher@schooliedu.local", "phon

GET /api/users/{id}

Purpose Get a user by ID.
Auth user.view
Response User profile.
Tables Reads users, user_roles, roles.
Request

```
Path: id UUID.
```

PATCH /api/users/{id}

Purpose Update profile fields.
Auth user.update
Response Updated user profile.
Tables Updates users.
Request

```
{"firstName":"Demo","lastName":"Teacher","phone":"+919999999999","avatarUrl":null}
```

PATCH /api/users/{id}/status

Purpose Activate, deactivate, or suspend a user.
Auth user.deactivate
Response Updated user profile.
Tables Updates users.status and users.is_active.
Request

```
{"status":"active","isActive":true}
```

POST /api/users/{id}/roles

Purpose Replace assigned roles.
Auth user.update
Response Updated user roles.
Tables Deletes/inserts user_roles.
Request

```
{"roles":["teacher"]}
```

People, Availability, And Assignments

Manage teacher/student profile views, weekly teacher availability, blocked dates, and teacher-student assignment prerequisites for scheduling.

Endpoint Catalogue

Method	Path	Permission
GET	/api/teachers	teacher.view
GET	/api/teachers/{id}	teacher.view
GET	/api/students	student.view
GET	/api/students/{id}	student.view
GET	/api/teacher-student-assignments	teacher.view/student.view scope
POST	/api/teacher-student-assignments	assignment create permission
PATCH	/api/teacher-student-assignments/{id}/status	assignment update permission
GET	/api/teachers/{id}/availability	teacher.view
POST	/api/teachers/{id}/availability	teacher.update
POST	/api/teachers/{id}/unavailable-dates	teacher.update
DELETE	/api/teachers/{id}/unavailable-dates/{dateId}	teacher.update

Detailed Endpoint Specs

GET	/api/teachers
Purpose	List teacher users.
Auth	teacher.view
Response	Paginated teachers.
Tables	Reads users and roles.
Request	
	Query: page, limit, search, status.

GET /api/teachers/{id}

Purpose Get teacher profile.
Auth teacher.view
Response Teacher profile.
Tables Reads users and roles.
Request

Path: id UUID.

GET /api/students

Purpose List student users.
Auth student.view
Response Paginated students.
Tables Reads users and roles.
Request

Query: page, limit, search, status.

GET /api/students/{id}

Purpose Get student profile.
Auth student.view
Response Student profile.
Tables Reads users and roles.
Request

Path: id UUID.

GET /api/teacher-student-assignments

Purpose List teacher-student links.
Auth teacher.view/student.view scope
Response Paginated assignments.
Tables Reads teacher_student_assignments and users.
Request

Query: page, limit, teacherId, studentId, status.

POST /api/teacher-student-assignments

Purpose Create an active teacher-student link.
Auth assignment create permission
Response Created assignment.
Tables Inserts teacher_student_assignments.
Notes Class scheduling requires this active link.

Request

```
{"teacherId":"<teacher uuid>","studentId":"<student uuid>","notes":"Initial assignment"}
```

PATCH /api/teacher-student-assignments/{id}/status

Purpose Activate/deactivate a link.
Auth assignment update permission
Response Updated assignment.
Tables Updates teacher_student_assignments.

Request

```
{"status":"active"}
```

GET /api/teachers/{id}/availability

Purpose List weekly slots and blocked dates.
Auth teacher.view
Response Availability bundle.
Tables Reads teacher_availability and teacher_unavailable_dates.

Request

Path: id UUID.

POST /api/teachers/{id}/availability

Purpose Create/update a weekly availability slot.
Auth teacher.update
Response Availability record.
Tables Inserts/updates teacher_availability.

Request

```
{"dayOfWeek":"monday","startTime":"16:00","endTime":"20:00","timezone":"Asia/Kolkata","isActive":true}
```

POST /api/teachers/{id}/unavailable-dates

Purpose Block a date/time window.
Auth teacher.update
Response Blocked-date record.
Tables Inserts teacher_unavailable_dates.

Request

```
{"unavailableDate": "2026-06-10", "startTime": "16:00", "endTime": "18:00", "reason": "Personal leave"}
```

DELETE /api/teachers/{id}/unavailable-dates/{dateId}

Purpose Remove a blocked date.
Auth teacher.update
Response Delete confirmation.
Tables Deletes teacher_unavailable_dates.

Request

Path: id UUID, dateId UUID.

Class Scheduling And Cancellation

Schedule teacher-student classes, create Zoom metadata, expose calendar data, and manage cancellation requests.

Endpoint Catalogue

Method	Path	Permission
POST	/api/classes/check-conflicts	class.create
GET	/api/classes	class.view
POST	/api/classes	class.create
GET	/api/classes/{id}	class.view
PATCH	/api/classes/{id}	class.update
POST	/api/classes/{id}/cancel	class.cancel
POST	/api/classes/{id}/reschedule	class.reschedule
POST	/api/classes/{id}/join	class.join
GET	/api/calendar/classes	class.view
GET	/api/classes/{id}/ics	class.view
GET	/api/classes/{id}/cancel-requests	class.view
POST	/api/classes/{id}/cancel-requests	class.view
GET	/api/class-cancellation-requests	class.view
PATCH	/api/class-cancellation-requests/{id}/status	class.cancel for approve/reject

Detailed Endpoint Specs

POST

/api/classes/check-conflicts

Purpose

Auth

Response

Tables

Request

Validate a proposed class time before creation.

class.create

hasConflicts plus conflict list.

Reads assignments, availability, blocked dates, classes, class_participants.

```
{"teacherId":"<teacher uuid>","studentId":"<student uuid>","startTime":"2026-06-10T10:00:00.000Z","durationMinutes":60,"timezone":"Asia/Kolkata"}
```

GET

/api/classes

Purpose

List classes scoped to current user.

Auth

class.view

Response

Paginated class list.

Tables

Reads classes, class_participants, users, zoom_meetings.

Request

Query: page, limit, status, teacherId, studentId, from, to.

POST

/api/classes

Purpose

Schedule a class for one teacher and one student.

Auth

class.create

Response

Created class with participant and Zoom metadata.

Tables

Inserts classes, class_participants, class_attendance, zoom_meetings.

Notes

Auto-creates Zoom when Zoom config is present.

Request

```
{"teacherId":"<teacher uuid>","studentId":"<student uuid>","title":"Math class","startTime":"2026-06-10T10:00:00.000Z","durationMinutes":60,"timezone":"Asia/Kolkata","notes":"Optional note","overrideConflicts":false}
```

GET

/api/classes/{id}

Purpose

Get class details.

Auth

class.view

Response

Class detail.

Tables

Reads classes, participants, users, zoom_meetings.

Request

Path: id UUID.

PATCH

/api/classes/{id}

Purpose

Update title/notes.

Auth

class.update

Response

Updated class.

Tables

Updates classes.

Request

```
{"title":"Updated title","notes":"Updated note"}
```

POST /api/classes/{id}/cancel

Purpose Directly cancel a class.
Auth class.cancel
Response Cancelled class.
Tables Updates classes.status, cancelled_at, cancellation_reason.

Request

```
{"reason": "Operational cancellation reason"}
```

POST /api/classes/{id}/reschedule

Purpose Move class to a new time after conflict checks.
Auth class.reschedule
Response Rescheduled class.
Tables Updates classes start/end/duration/timezone/status.

Request

```
{"startTime": "2026-06-11T10:00:00.000Z", "durationMinutes": 60, "timezone": "Asia/Kolkata", "overrideConflicts": false}
```

POST /api/classes/{id}/join

Purpose Return Zoom join/start payload.
Auth class.join
Response Meeting provider payload.
Tables Reads classes and zoom_meetings.
Notes Teacher/admin can receive returnUrl; student receives joinUrl.

Request

Path: id UUID.

GET /api/calendar/classes

Purpose Calendar-oriented class list.
Auth class.view
Response Class list.
Tables Reads same tables as /api/classes.

Request

Query: from, to, teacherId, studentId.

GET /api/classes/{id}/ics**Purpose** Download ICS event file.**Auth** class.view**Response** text/calendar response.**Tables** Reads classes.**Request**

Path: id UUID.

GET /api/classes/{id}/cancel-requests**Purpose** List cancellation requests for one class.**Auth** class.view**Response** Paginated requests.**Tables** Reads class_cancellation_requests, classes, users.**Request**

Path id. Query: page, limit, status.

POST /api/classes/{id}/cancel-requests**Purpose** Student/teacher cancellation request with reason.**Auth** class.view**Response** Created request.**Tables** Inserts class_cancellation_requests.**Request**

```
{"reason": "Student has an exam at the same time"}
```

GET /api/class-cancellation-requests**Purpose** Admin/support cancellation queue.**Auth** class.view**Response** Paginated requests.**Tables** Reads class_cancellation_requests, classes, users.**Request**

Query: page, limit, status, classId, requestedByUserId.

PATCH /api/class-cancellation-requests/{id}/status	
Purpose	Approve, reject, or withdraw request.
Auth	class.cancel for approve/reject
Response	Updated request.
Tables	Updates class_cancellation_requests. If approved, updates classes to cancelled.
Request	
{ "status": "approved", "adminNote": "Approved by admin" }	

Attendance

Mark official attendance while preserving Zoom join/leave evidence.

Endpoint Catalogue

Method	Path	Permission
GET	/api/attendance	attendance.view
GET	/api/classes/{id}/attendance	attendance.view
POST	/api/attendance/mark	attendance.mark
PATCH	/api/attendance/{id}	attendance.mark; override users use attendance.override

Detailed Endpoint Specs

GET	/api/attendance
Purpose	List attendance records.
Auth	attendance.view
Response	Paginated attendance records with Zoom evidence summary.
Tables	Reads class_attendance, classes, users, zoom_attendance_events.
Request	Query: page, limit, classId, teacherId, studentId, status, from, to.

GET	/api/classes/{id}/attendance
Purpose	List attendance for one class.
Auth	attendance.view
Response	Attendance records.
Tables	Reads class_attendance, classes, users, zoom_attendance_events.
Request	Path: id UUID.

POST

/api/attendance/mark

Purpose	Mark attendance.
Auth	attendance.mark
Response	Marked attendance record.
Tables	Upserts class_attendance and updates class_participants.attendance_status.

Request

```
{"classId":"<class uuid>","studentId":"<student uuid>","status":"present","teacherNotes":"Joined on time","zoomJoinTime":"2026-06-10T10:00:00.000Z","zoomLeaveTime":"2026-06-10T10:57:00.000Z","totalZoomMinutes":57}
```

PATCH

/api/attendance/{id}

Purpose	Update attendance.
Auth	attendance.mark; override users use attendance.override
Response	Updated attendance record.
Tables	Updates class_attendance and class_participants.

Request

```
{"status":"late","teacherNotes":"Joined late","totalZoomMinutes":42}
```

Zoom

Create and inspect Zoom meetings, handle Zoom webhooks, and prepare embedded meeting signatures.

Endpoint Catalogue

Method	Path	Permission
POST	/api/zoom/meetings	class.create/Zoom route guard
GET	/api/zoom/meetings/{id}	class.view
POST	/api/zoom/webhook	Zoom signature verification
POST	/api/classes/{id}/zoom/signature	class.join

Detailed Endpoint Specs

POST**/api/zoom/meetings**

Purpose Create or recreate Zoom meeting for class.

Auth class.create/Zoom route guard

Response Zoom meeting metadata.

Tables Reads classes. Inserts/updates zoom_meetings.

Notes Requires Zoom Server-to-Server OAuth env vars.

Request

```
{"classId":"<class uuid>"}
```

GET**/api/zoom/meetings/{id}**

Purpose Get local Zoom metadata.

Auth class.view

Response Zoom meeting record.

Tables Reads zoom_meetings and classes.

Request

Path: id UUID.

POST**/api/zoom/webhook**

Purpose Receive Zoom url_validation and join/leave events.

Auth Zoom signature verification

Response Webhook acknowledgement.

Tables Inserts zoom_attendance_events; may update zoom_meetings.

Request

Zoom webhook payload.

POST /api/classes/{id}/zoom/signature	
Purpose	Generate Meeting SDK signature.
Auth	class.join
Response	Meeting SDK signature payload.
Tables	Reads classes and zoom_meetings.
Notes	Requires ZOOM_MEETING_SDK_KEY and ZOOM_MEETING_SDK_SECRET.
Request	{"role":0}

Email Templates And Notification Manager

Configure reusable HTML mail templates and rules for when notifications should be sent.

Endpoint Catalogue

Method	Path	Permission
GET	/api/email-templates	settings.view
POST	/api/email-templates	settings.update
GET	/api/email-templates/{id}	settings.view
PATCH	/api/email-templates/{id}	settings.update
PATCH	/api/email-templates/{id}/status	settings.update
GET	/api/notification-rules	settings.view
POST	/api/notification-rules	settings.update
PATCH	/api/notification-rules/{id}	settings.update
PATCH	/api/notification-rules/{id}/status	settings.update
GET	/api/notification-delivery-logs	settings.view

Detailed Endpoint Specs

GET

/api/email-templates

Purpose

Auth

Response

Tables

Request

List email templates.

settings.view

Paginated templates.

Reads email_templates.

Query: page, limit, search, isActive.

POST

/api/email-templates

Purpose

Auth

Response

Tables

Request

Create HTML email template.

settings.update

Created template.

Inserts email_templates.

```
{"key": "class.scheduled.default", "name": "Class Scheduled", "subject": "Your class is scheduled", "htmlBody": "<h1>Hello {{studentName}}</h1>", "textBody": "Hello {{studentName}}", "availableVariables": ["studentName", "teacherName", "startTime", "joinUrl"], "isActive": true}
```

GET /api/email-templates/{id}

Purpose Get one template.
Auth settings.view
Response Template detail.
Tables Reads email_templates.
Request

Path: id UUID.

PATCH /api/email-templates/{id}

Purpose Update template fields.
Auth settings.update
Response Updated template.
Tables Updates email_templates.
Request

```
{"subject": "Updated subject", "htmlBody": "<p>Updated HTML</p>"}
```

PATCH /api/email-templates/{id}/status

Purpose Enable/disable template.
Auth settings.update
Response Updated template status.
Tables Updates email_templates.is_active.
Request

```
{"isActive": true}
```

GET /api/notification-rules

Purpose List notification rules.
Auth settings.view
Response Paginated rules.
Tables Reads notification_rules and email_templates.
Request

Query: page, limit, eventKey, channel, recipientRole, isEnabled.

POST /api/notification-rules

Purpose Create notification rule.
Auth settings.update
Response Created rule.
Tables Inserts notification_rules.

Request

```
{"eventKey":"class.scheduled","channel":"email","recipientRole":"student","emailTemplateId":"<template uuid>","isEnabled":true,"conditions":{}}
```

PATCH /api/notification-rules/{id}

Purpose Update notification rule.
Auth settings.update
Response Updated rule.
Tables Updates notification_rules.

Request

```
{"eventKey":"class.cancelled","emailTemplateId":"<template uuid>","conditions":{}}
```

PATCH /api/notification-rules/{id}/status

Purpose Enable/disable notification rule.
Auth settings.update
Response Updated rule status.
Tables Updates notification_rules.is_enabled.

Request

```
{"isEnabled":false}
```

GET /api/notification-delivery-logs

Purpose List delivery logs.
Auth settings.view
Response Paginated delivery logs.
Tables Reads notification_delivery_logs.
Notes Actual SMTP/provider sending is future work.

Request

Query: page, limit, eventKey, channel, status, recipientUserId.

Roadmap Boundaries

The following database schemas already exist but public API modules are still pending: homework, credits ledger, support tickets, certificates, reports, settings, and audit logs. They should be documented in this guide as soon as each API slice is implemented.

Current Phase 1 Boundaries

- Parent role is excluded from Phase 1.
- Payments are excluded from Phase 1.
- Real frontend Zoom SDK integration waits on Meeting SDK key/secret.
- Real SMTP/provider email sending is not yet implemented.
- Cloudinary/media upload integration is deferred.